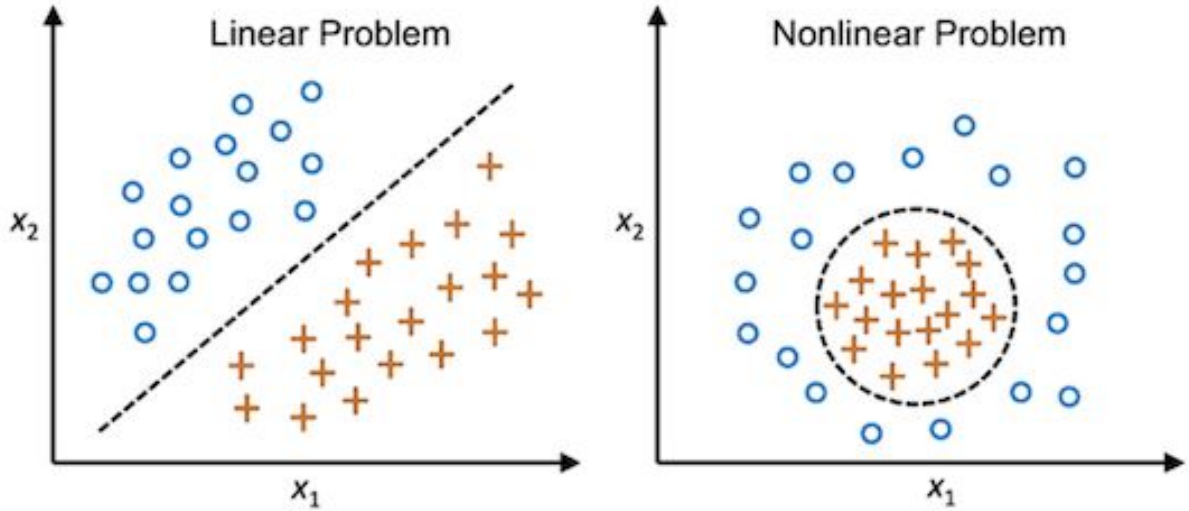
# Support Vector Machines (SVM)

Jelena Jovanovic

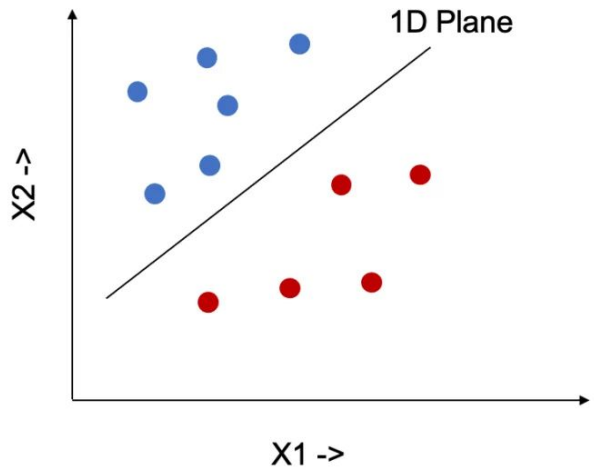
# Pre-intro: linearni i nelinearni zadaci klasifikacije

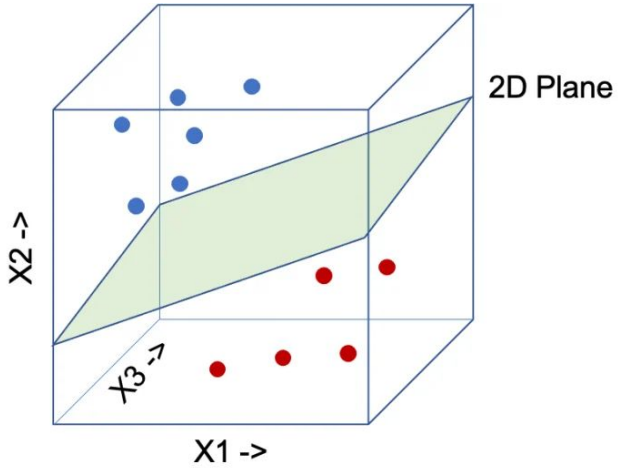



Izvor slike: https://www.analyticsvidhya.com/blog/2021/05/support-vector-machines/

# SVM: Osnovna ideja linearnog SVM-a

Pronalaženje najbolje moguće “linije razgraničenja” (hiper-ravni) za razdvajanje dve klase podataka u visokodimenzionalnom prostoru

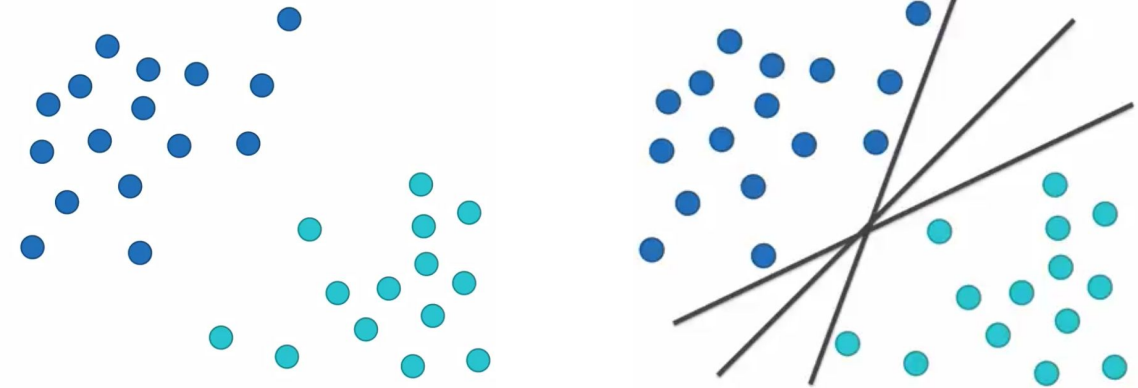



Izvor slike: https://adeveloperdiary.com/data-science/machine-learning/support-vector-machines-for-beginners-linear-svm/

# SVM: Optimalna hiperravan

Za bilo koji dati problem klasifikacije gde su podaci *linearno razdvojivi*, postoji mnogo mogućih hiperravni koje bi mogle razdvojiti klase.

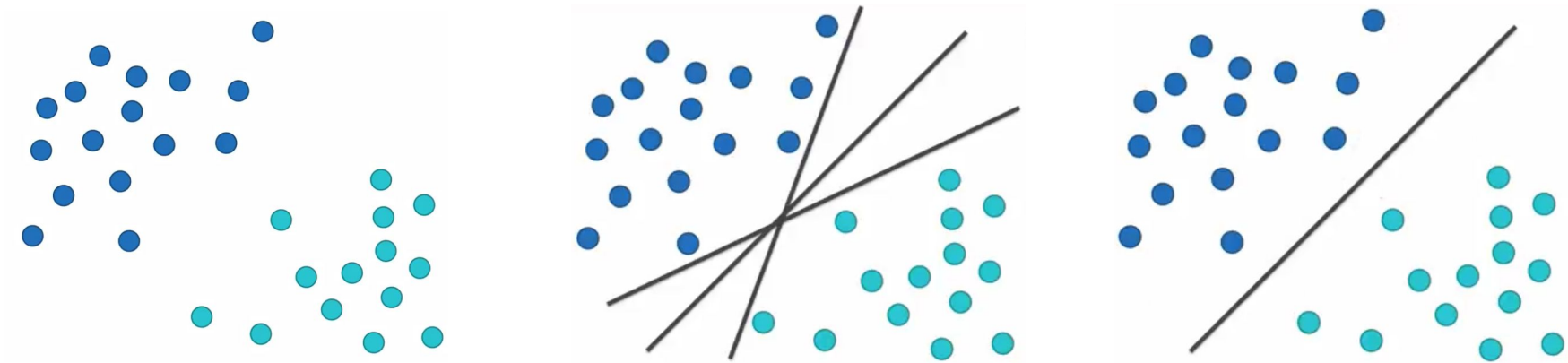


Izvor slika: https://www.youtube.com/watch?v=N1vOgolbjSc

# SVM: Optimalna hiperravan

Za bilo koji dati problem klasifikacije gde su podaci *linearno razdvojivi*, postoji mnogo mogućih hiperravni koje bi mogle razdvojiti klase.

SVM algoritam traži ***optimalnu hiperravan***, onu koja maksimalno razdvaja pripadnike dve klase



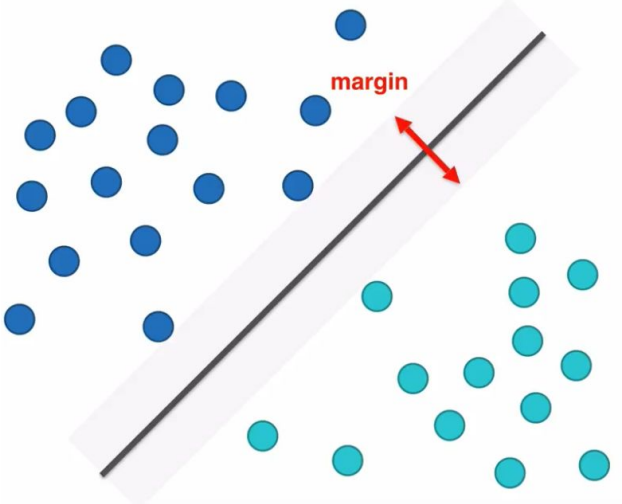
Izvor slika: https://www.youtube.com/watch?v=N1vOgolbjSc

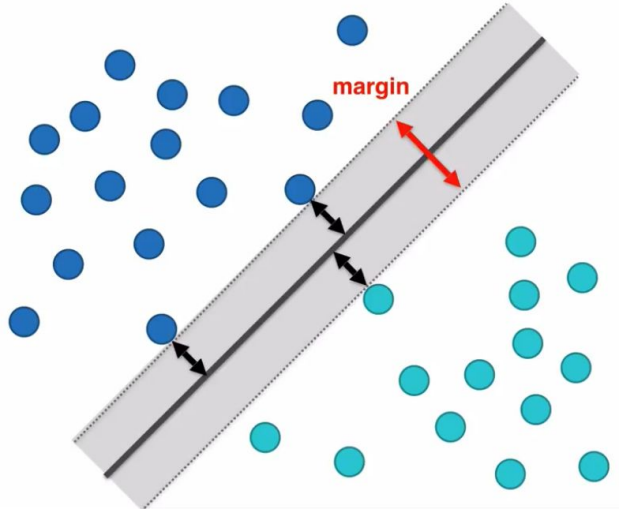
# SVM: Optimalna hiperravan

Optimalna hiperravan je ona koja postiže maksimalnu ***marginu***.

Margina je rastojanje između hiperravni i najbliže tačke podataka iz svake od klasa

Cilj SVM-a je da pronađe hiperravan koja ovu marginu čini što je moguće širom.




Izvor slika: https://www.youtube.com/watch?v=N1vOgolbjSc

# Linearna funkcija i maksimizacija širine margine

Linearni SVM je zasnovan na linearnoj funkciji:

$$z = b + w^T \cdot x$$

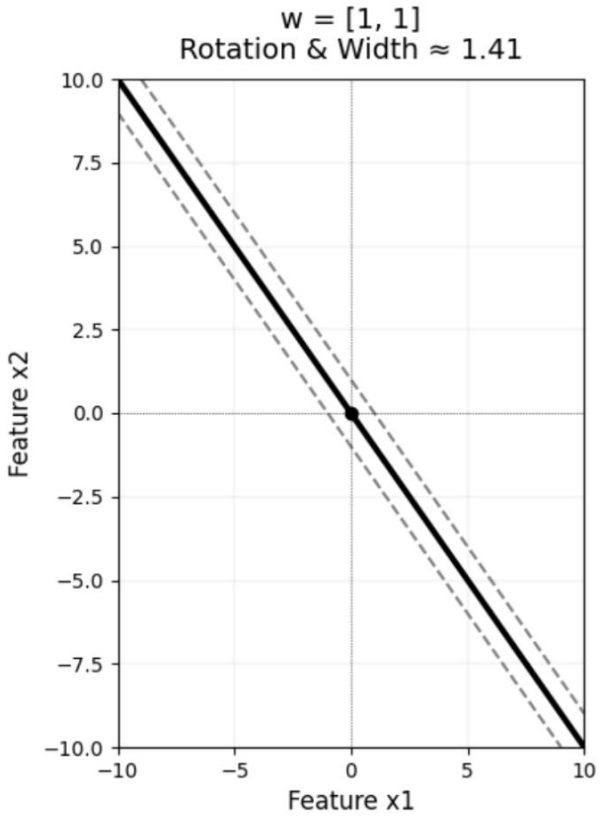
Cilj je odrediti parametre ***w*** i ***b*** tako da se maksimizira širina margine, definisana kao $2/\|w\|$

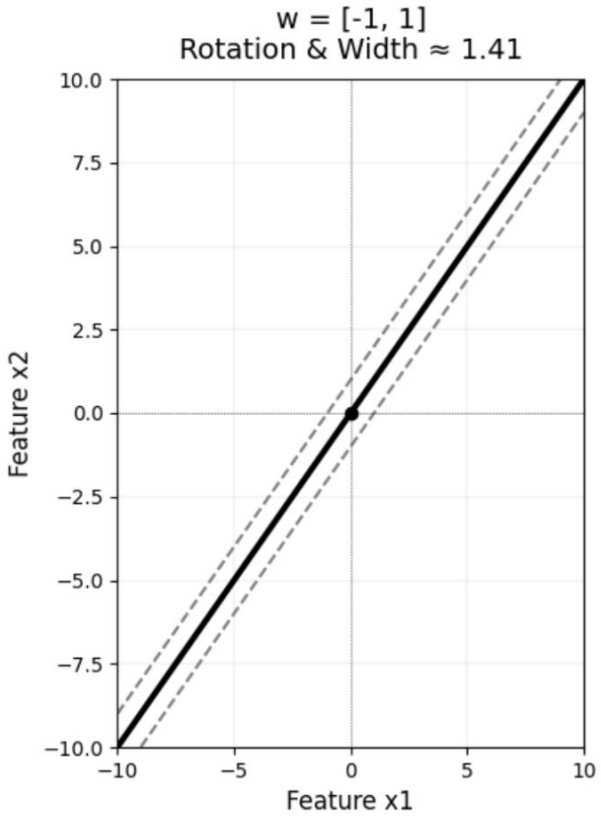
Parametar ***w*** definiše “nagib” hiperravni i širinu margine

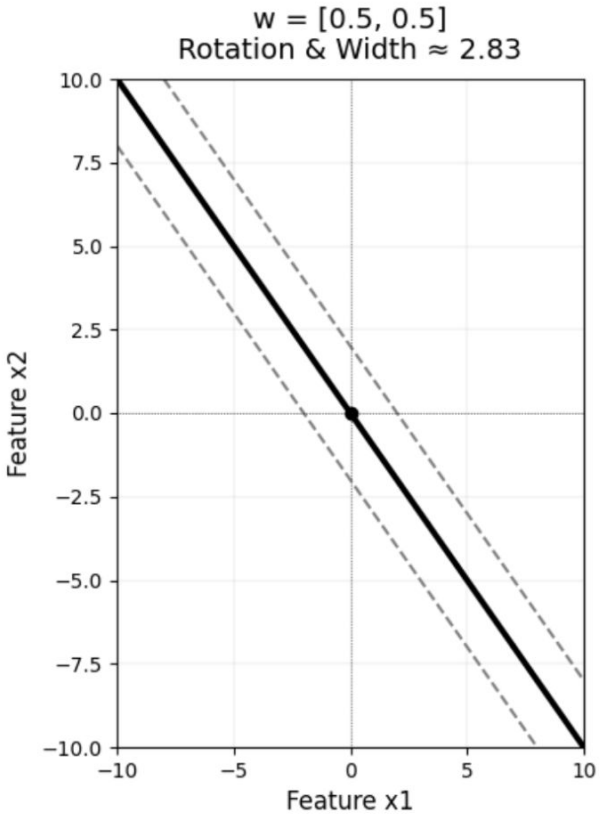
Parametar ***b*** određuje da li je hiperravan pomerena (translirana) malo levo / desno

# Efekat težina (w) na položaj i nagib hiperravni

Fiksirano
b = 0

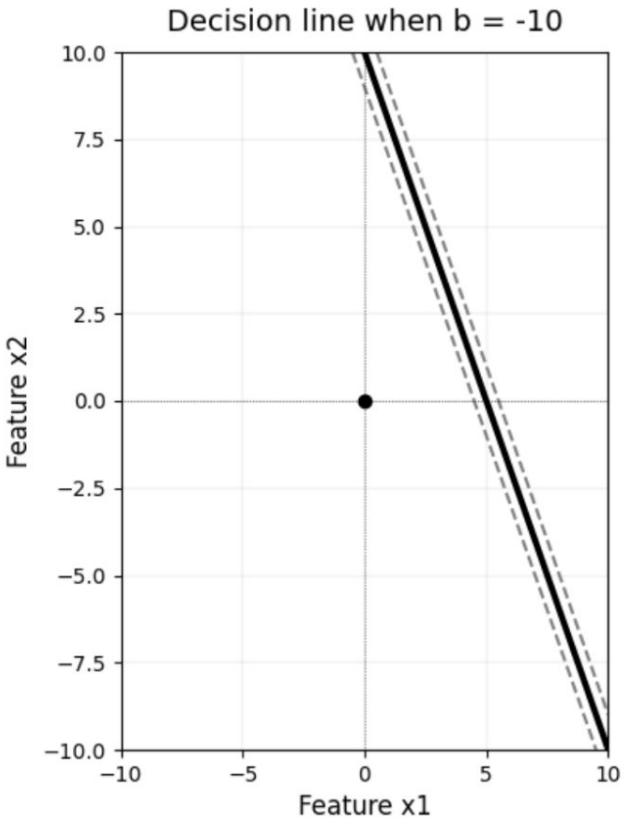


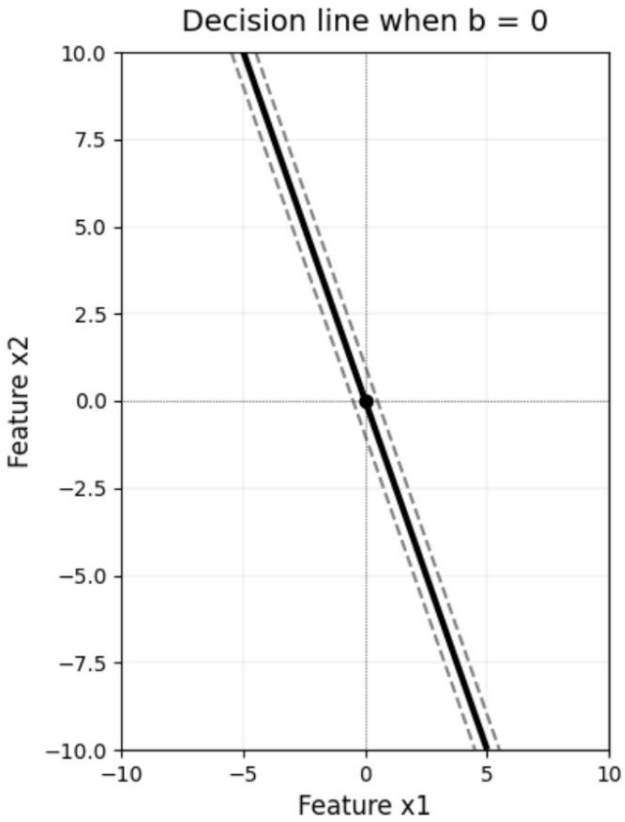



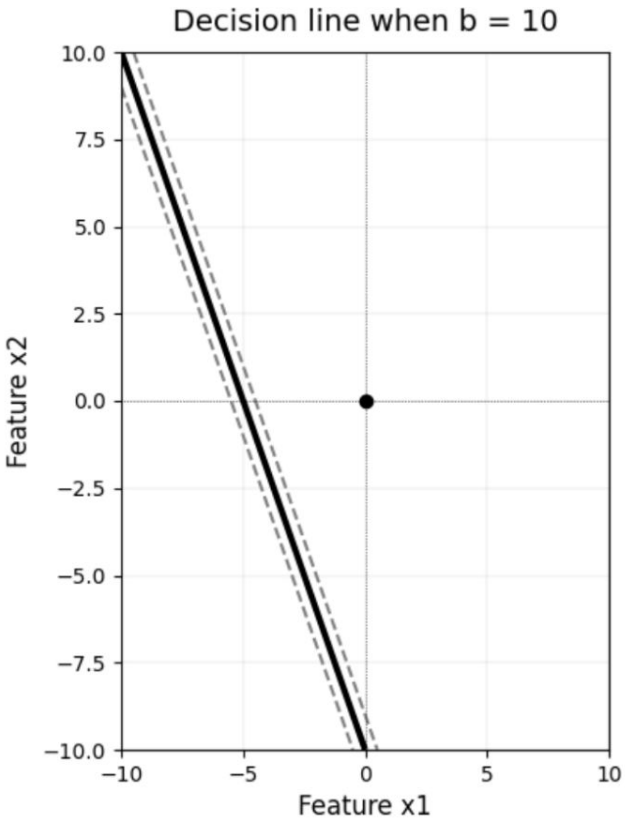



# Efekat slobodnog člana (b) na poziciju hiperravni

Fiksirano
w = [2, 1]



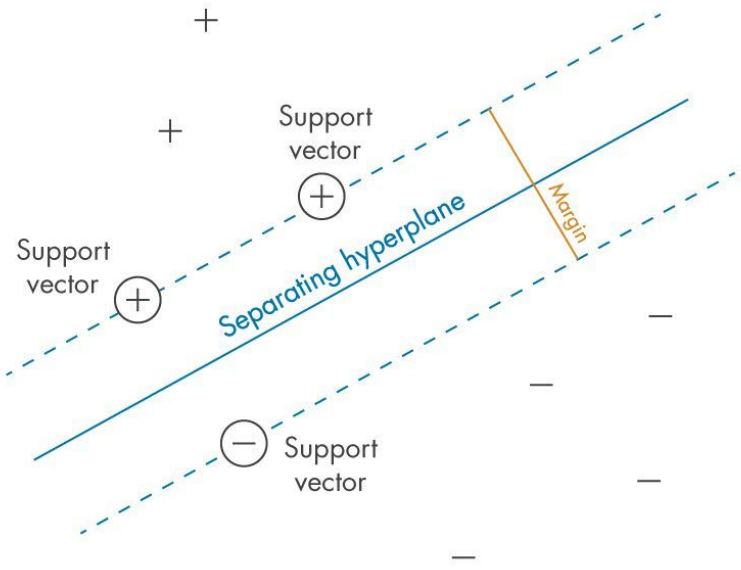



# Uloga potpornih vektora

Opservacije iz svake klase koje se nalaze na margini optimalne hiperravni nazivaju se ***potporni vektori***

Ove opservacije su jedine bitne za određivanje parametara *w* i *b* linearne funkcije, odnosno, definisanje položaja i orijentacije optimalne hiperravni

Sve ostale observacije se mogu ukloniti, a optimalna hiperravan se ne bi promenila

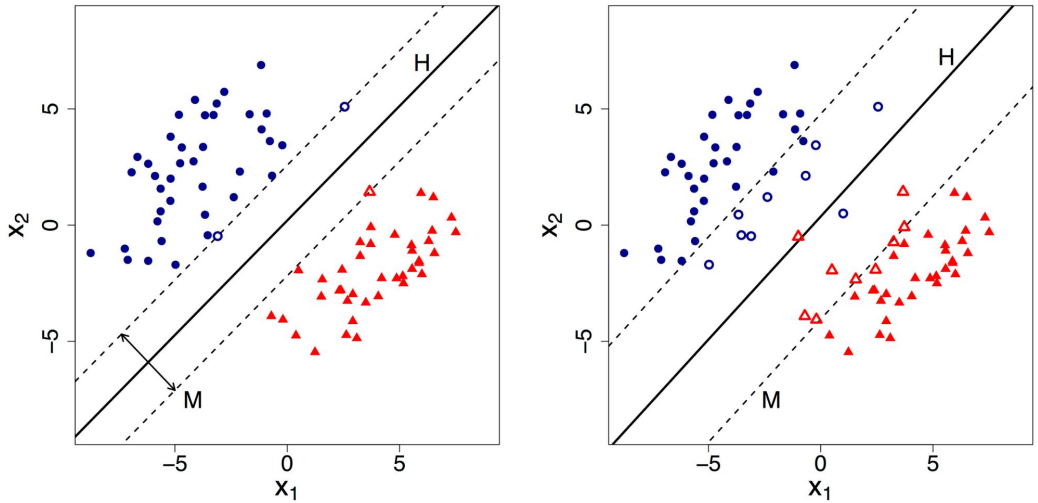



Izvor slike: https://www.mathworks.com/discovery/support-vector-machine.html

# SVM: Mekane margine

Meka margina (soft margin) SVM-a ublažava strogi zahtev za savršenim razdvajanjem klasa i time smanjuje rizik od overfitting-a

Dozvoljava nekoliko pogrešnih klasifikacija u trening podacima kako bi se postigla šira, robusnija margina




Izvor slike: https://www.analyticsvidhya.com/blog/2021/04/insight-into-svm-support-vector-machine-along-with-code/

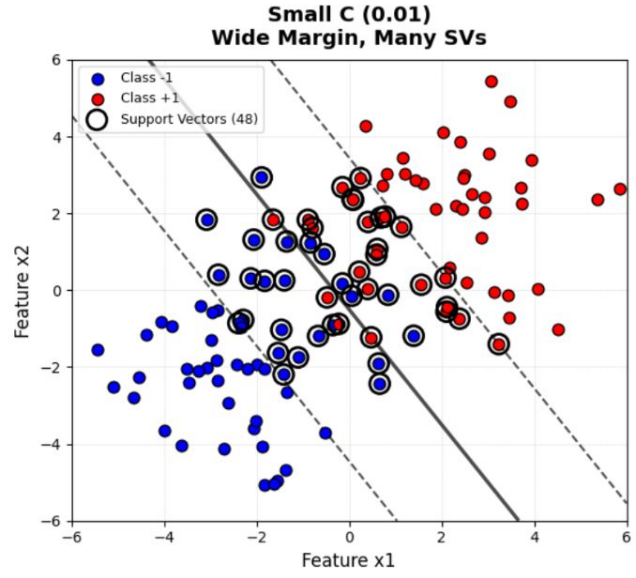
# SVM: Mekane margine

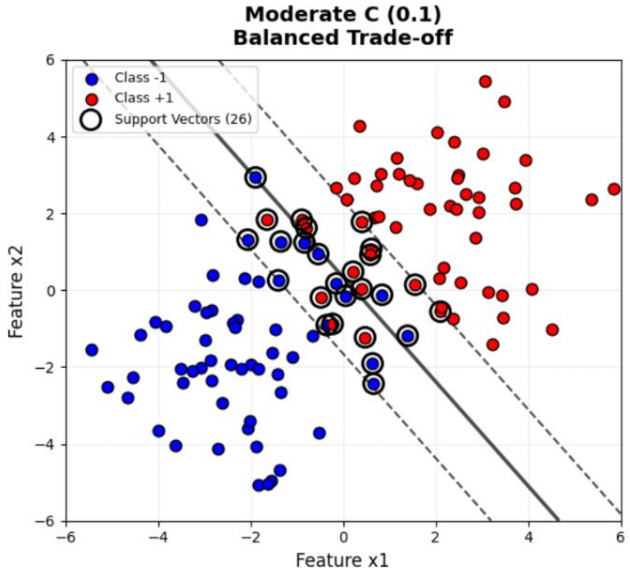
Ovo ‘slabljenje’ margina kontroliše hiperparametar C:

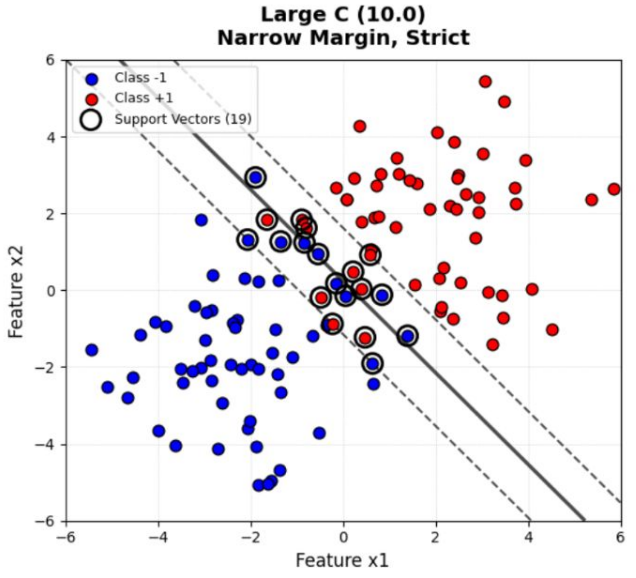
- **Veliko C** nameće usku marginu i jače kažnjava pogrešne klasifikacije, što znači bolje prilagođavanje podacima za obuku, ali i rizik overfitting-a
- **Malo C** omogućava široku marginu i više pogrešnih klasifikacija, ali i manji rizik od overfitting-a

# SVM: Mekane margine

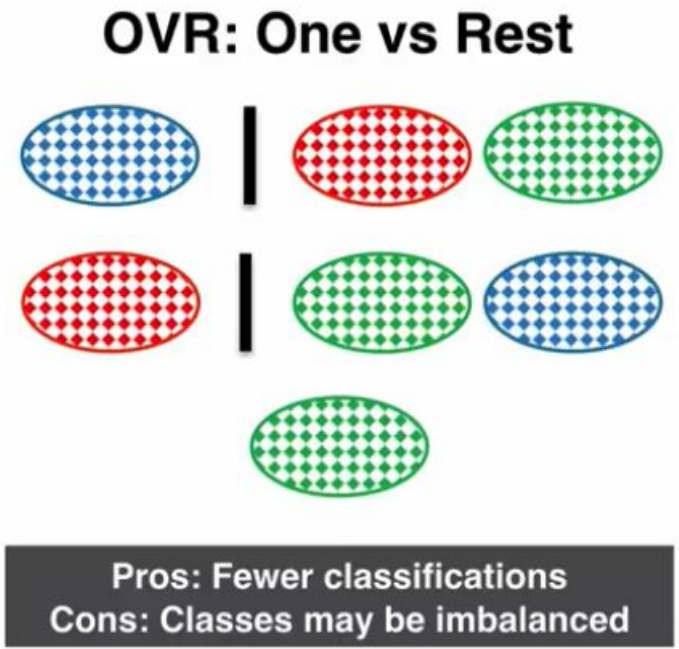
Efekat hiperparametra C na širinu margine i broj potpornih vektora

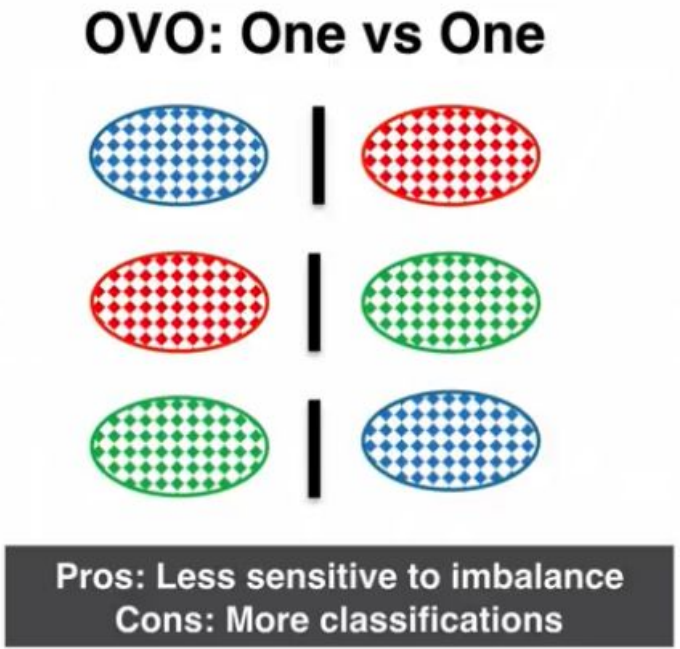


# Višeklasna klasifikacija




Strategija koju primenjuje sklearn-ova implementacija (SVC klasa)

Izvor slike: https://www.youtube.com/watch?v=N1vOgolbjSc

# Višeklasna klasifikacija: One vs One

Klase iz primera: Other, Functional, Non-functional

3 Binarna klasifikatora:

- 0 (Other) vs 1 (Functional)
- 0 (Other) vs 2 (Non-Functional)
- 1 (Functional) vs 2 (Non-Functional)

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Prediction scores:

-1.2679

-1.6188

-1.78029

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$$z = b + w^T \cdot x$$

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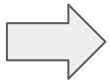
Prediction scores:

-1.2679

-1.6188

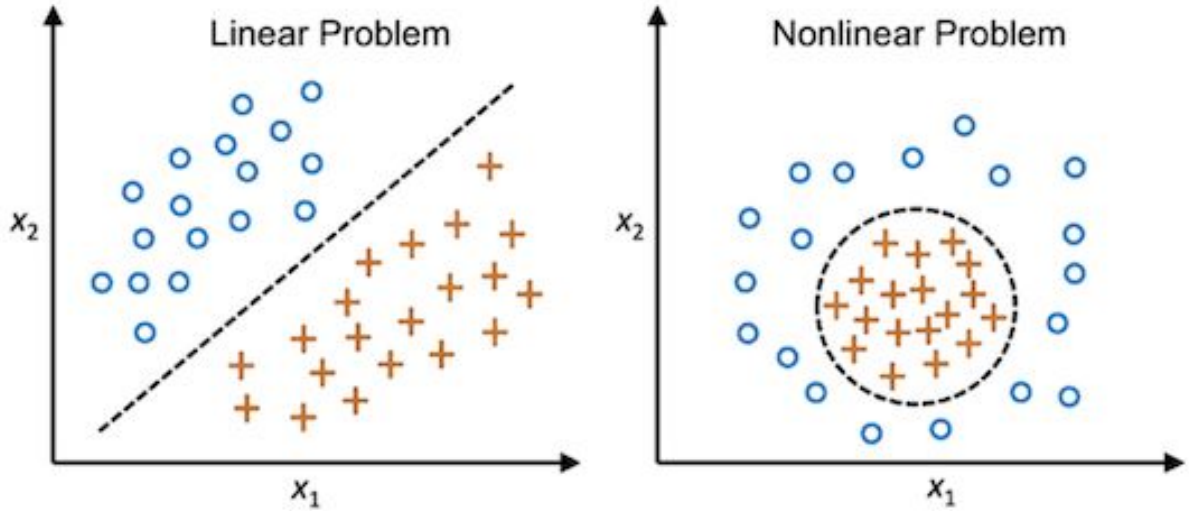
-1.78029

-1.2679 => 1 (Functional)
-1.6188 => 2 (Non-Functional)
-1.7802 => 2 (Non-Functional)



2 (Non-Functional)

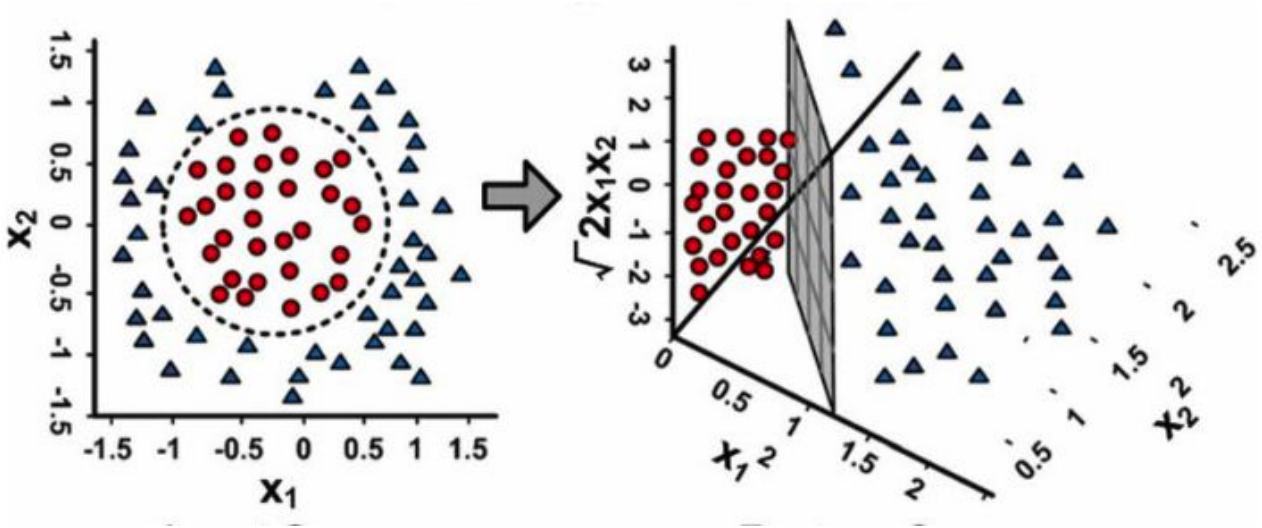
# SVM: Podaci koji nisu linearno razdvojivi




Izvor slike: https://www.analyticsvidhya.com/blog/2021/05/support-vector-machines/

# SVM: Podaci koji nisu linearno razdvojivi

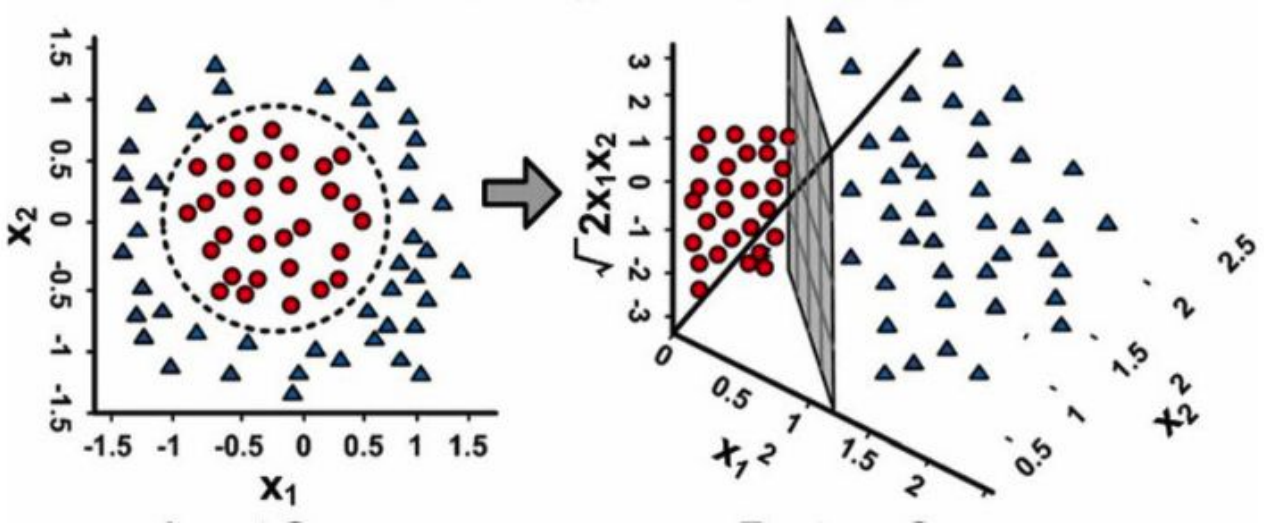
Kernel funkcije za transformaciju ulaznih podataka u linearno razdvojivi višedimenzionalni prostor




Izvor slike: https://www.researchgate.net/figure/Motivation-behind-non-linear-SVM-classifier_fig2_272520997

# SVM: Podaci koji nisu linearno razdvojivi

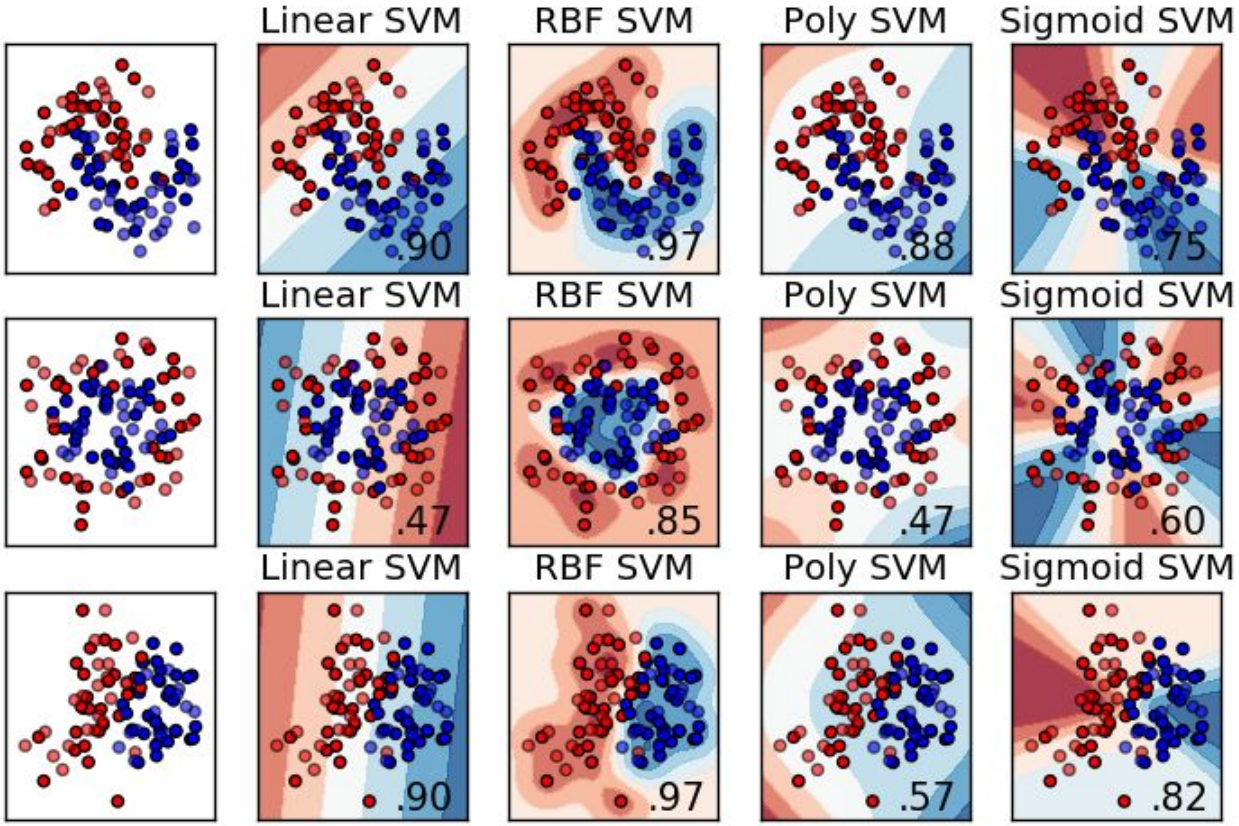
Kernel funkcije za transformaciju ulaznih podataka u linearno razdvojivi višedimenzionalni prostor




Najčešće korišćene kernel funkcije:
‘poly’,
‘rbf’,
‘sigmoid’

Izvor slike: https://www.researchgate.net/figure/Motivation-behind-non-linear-SVM-classifier_fig2_272520997

# Ilustracija primene različitih kernela




https://gist.github.com/WittmannF/60680723ed8dd0cb993051a7448f7805